Project Title

BookNest: A Hotel Management System

Course Title: Database Management System Lab
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1. Introduction

BookNest: A Hotel Management System is a web-based application that automates hotel operations using a frontend, backend, and MySQL database. It manages reservations, guests, employees, payments, housekeeping, maintenance, inventory, and reporting.

The system supports role-based access for Owner, Manager, Receptionist, Housekeeping Staff, and Maintenance Staff to ensure secure and efficient hotel management.

2. Problem Statement

Many hotels still rely on manual record keeping or disconnected software systems, resulting in:

- Data redundancy
- Booking conflicts
- Slow customer service
- Human errors
- Difficulty managing employees and hotel resources
- Limited reporting capabilities

BookNest aims to solve these problems by providing a centralized database-driven hotel management system that enables efficient CRUD operations, secure user authentication, and real-time hotel management.

3. Objectives

The main objectives of BookNest are:

- Develop a complete web-based Hotel Management System.
- Design a normalized MySQL database (1NF, 2NF, and 3NF).
- Implement complete CRUD operations for all modules.
- Connect the React frontend with the Node.js backend and MySQL database.
- Ensure secure authentication with role-based access control.
- Improve hotel operational efficiency through automation.
- Generate reports and analytics for management.
- Maintain data integrity using primary keys, foreign keys, constraints, and triggers.

4. Technologies System Modules

Frontend

- HTML5
- Vite
- Tailwind CSS / CSS3
- JavaScript
- React.js

Backend

- Node.js /Express.js

Database

- MySQL

Server

- XAMPP / Local MySQL Server

Development Tools

- Visual Studio Cod
- Xcode
- Netbeans
- Git & GitHub

Authentication & Security

- User Login
- User Logout
- Password Management
- Role-Based Access Control
- Activity Logs
- Data Backup & Recovery

User Management

- Owner Management
- Manager Management
- Employee Account Management
- Role Assignment
- Permission Management

Room Management

- Add/Edit/Delete Rooms
- Room Availability
- Room Booking
- Room Reservation
- Room Pricing
- Room Status
- Room Type Management
- Room Amenities

Guest Management

- Guest Registration
- Guest Profile
- Guest History
- Guest Identification
- Emergency Contacts
- Guest Feedback

Reservation Management

- Create Reservation
- Modify Reservation
- Cancel Reservation
- Booking Confirmation
- Walk-in Booking
- Advance Booking

Check-in & Check-out

- Guest Check-in
- Guest Check-out
- Room Allocation
- Booking Verification

Payment Management

- Cash Payment
- Card Payment
- Mobile Banking
- Online Payment
- Invoice Generation
- Refund Management
- Payment History

Employee Management

- Employee Registration
- Attendance
- Salary Management
- Work Schedule
- Performance Records

Housekeeping Management

- Cleaning Schedule
- Room Cleaning Status
- Lost & Found
- Damage Reports

Maintenance Management

- Maintenance Requests
- Repair Status
- Equipment Maintenance
- Repair Cost Records

Inventory Management

- Room Supplies
- Restaurant Inventory
- Linen Management
- Stock Control
- Purchase Records
- Low Stock Alerts

Hotel Services

- Room Service
- Food & Beverage
- Laundry
- Wake-up Call
- Airport Pickup
- Additional Services

Reporting System

- Booking Reports
- Occupancy Reports
- Revenue Reports
- Payment Reports
- Salary Reports
- Attendance Reports
- Service Reports
- Maintenance Reports

5. User Roles

Owner

- Complete control of the system
- Create, update, or remove Managers
- Configure system settings
- Access all reports
- Manage all hotel operations

Manager (Administrator)

- Manage hotel operations
- Manage employees
- Approve discounts and refunds
- Manage salaries
- View reports
- Manage rooms and services

Receptionist

- Register guests
- Manage bookings
- Check-in/Check-out
- Receive payments
- Generate invoices

Housekeeping Staff

- View assigned rooms
- Update cleaning status
- Report damages
- Attendance tracking

Maintenance Staff

- View maintenance requests
- Update repair status
- Record repair costs
- Complete maintenance jobs

6. Database Tables

The proposed database contains the following major tables:

Table Name	**Purpose**
Users	User login information
Roles	User roles and permissions
Employees	Employee information
Attendance	Employee attendance
Guests	Guest details
Rooms	Hotel rooms
Room Types	Room categories
Bookings	Reservation records
Check-ins	Guest check-in records
Check-outs	Guest check-out records
Payments	Payment information
Services	Hotel services
Service Requests	Guest service requests
Housekeeping	Cleaning records
Maintenance	Maintenance records
Inventory	Hotel inventory
Salaries	Employee salary records
Feedback	Guest feedback

Table Name	Purpose
Reports	Generated reports

7. CRUD Operations

The system supports complete CRUD functionality.

Create

- Register guests
- Add rooms
- Create bookings
- Add employees
- Record payments
- Create maintenance requests

Read

- Search rooms
- View guest information
- View booking history
- Display reports
- View employee records

Update

- Modify bookings
- Update guest profiles
- Update room status
- Edit employee information
- Update inventory

Delete

- Remove room records
- Delete cancelled bookings
- Remove employee records (authorized users)
- Delete outdated inventory records

8. Proposed Database Design

The database follows normalization principles (1NF, 2NF, and 3NF) and uses relational integrity through primary keys and foreign keys.

Example Table Structure

Table	Important Fields
Users	User_ID (PK), Username, Password, Role ID (FK), Employee_ID (FK), Status
Roles	Role_ID (PK), Role_Name
Employees	Employee_ID (PK), Name, Phone, Email, Department, Salary_ID (FK)
Guests	Guest_ID (PK), Name, Phone, Email, NID/Passport, Address
Room Types	RoomType_ID (PK), Type_Name, Base_Price
Rooms	Room_ID (PK), Room_Number, RoomType_ID (FK), Price, Status
Bookings	Booking_ID (PK), Guest_ID (FK), Room_ID (FK), Booking_Date, CheckIn_Date, CheckOut_Date, Status
Check-ins	CheckIn_ID (PK), Booking_ID (FK), CheckIn_Time
Check-outs	CheckOut_ID (PK), Booking_ID (FK), CheckOut_Time
Payments	Payment_ID (PK), Booking_ID (FK), Amount, Payment_Method, Payment_Date
Services	Service_ID (PK), Service_Name, Cost
Service Requests	Request_ID (PK), Guest_ID (FK), Service_ID (FK), Status
Housekeeping	Housekeeping_ID (PK), Room_ID (FK), Employee_ID (FK), Cleaning_Status
Maintenance	Maintenance_ID (PK), Room_ID (FK), Employee_ID (FK), Repair_Cost, Status
Inventory	Inventory_ID (PK), Item_Name, Quantity, Stock_Status
Attendance	Attendance_ID (PK), Employee_ID (FK), Date, Check_In, Check_Out
Salaries	Salary_ID (PK), Employee_ID (FK), Basic_Salary, Bonus
Feedback	Feedback_ID (PK), Guest_ID (FK), Rating, Comments
Reports	Report_ID (PK), Report_Type, Generated_Date

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9. Functional Requirements

The system shall allow users to:

- Login securely
- Manage user accounts
- Register guests
- Manage hotel rooms
- Handle reservations
- Perform guest check-in/check-out
- Process payments
- Manage employees
- Track attendance
- Handle housekeeping

- Process maintenance requests
- Manage inventory
- Generate invoices
- Generate hotel reports
- Search records efficiently
- Perform Create, Read, Update, and Delete (CRUD) operations according to user permissions

10. Non-Functional Requirements

- Responsive user interface
- Secure authentication and authorization
- Reliable database connectivity
- Fast data retrieval
- High data integrity
- Scalable database design
- Easy maintenance
- Role-based security
- Backup and recovery support
- User-friendly navigation

11. Expected Output

The final system will provide:

- A fully functional Hotel Management System
- Complete MySQL database integration
- Secure user authentication
- Role-based dashboard for different users
- Complete CRUD functionality
- Automated invoice generation
- Real-time room availability
- Reservation and payment management
- Employee and attendance management
- Housekeeping and maintenance tracking
- Inventory management
- Comprehensive reporting and analytics

12. Timeline

Week 1: Requirement Analysis, Literature Review & ER Diagram Design
Week 2: Relational Schema Design & Database Implementation
Week 3: Frontend Development & UI Integration
Week 4: Backend Development & API Development MySQL Integration
Week 5: CRUD Operations, Authentication, Testing, Debugging & Final Presentation

13. Conclusion

BookNest: A Hotel Management System is designed to provide a comprehensive and efficient solution for managing hotel operations. By integrating a React-based frontend, a Node.js backend, and a normalized MySQL database, the system ensures reliable data management, secure access control, and efficient hotel administration.